Md Ahsanul Haque, Ph.D.
MAT-120

Homework-3

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Instructions:

- Write your full name (first **and** last name) **inside** the provided box.
- This assignment contains 5 problems. It is **YOUR responsibility** to check that your copy has all 5 problems.
- Make sure to read the instructions for each problem carefully.
- Use correct mathematical notation.
- Write cohesively and follow a logical sequence of steps.
- Check that your answer can be read and understood by the grader.

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1. A water tank in the shape of an inverted right circular cone has a top radius of 4 feet and a height of 12 feet. Water is being pumped into the tank at a constant rate of 10 ft^3/min.
   (a) At what rate is the water level rising when the water in the tank is 6 feet deep?
   (b) At what rate is the wetted surface area of the cone changing at that same instant? (Recall that for a cone with radius r and height h, the lateral surface area is $A = \pi r\sqrt{r^2 + h^2}$).
   (c) At what rate is the volume of water changing with respect to the water height h when $h = 6$ feet?

2. Evaluate each limit using L'Hôpital's Rule (show all indeterminate forms before applying the rule):
   (a) $\lim_{x\to 0} \dfrac{e^{4x} - 1 - 4x}{x^2}$
   (b) $\lim_{x\to 0} \dfrac{1 - \cos(3x)}{x\tan(x)}$
   (c) $\lim_{x\to \infty} \dfrac{\ln(x^3 + 1)}{x}$
   (d) $\lim_{x\to 0^+} (x^2 \ln x)$

3. The concentration of a drug in a patient's bloodstream t hours after injection is given by $C(t) = 150e^{-0.12t}$, where $C(t)$ is measured in mg/L.
   (a) Find the rate at which the drug concentration is decreasing 2 hours after the injection.
   (b) How many hours will it take for the drug concentration to drop to 30 mg/L?
   (c) Find the average rate of change of the drug concentration over the first 4 hours.
   (d) Using L'Hôpital's Rule, evaluate $\lim_{t\to\infty} \dfrac{C(t)+t}{3t+2}$ and interpret its real-world meaning.

4. Consider the polynomial function $f(x) = x^4 - 4x^3 + 10$.
   (a) Find all critical points of $f(x)$.
   (b) Determine the open intervals on which $f(x)$ is increasing and decreasing. Show your work clearly using a sign chart for $f'(x)$.
   (c) Apply the First Derivative Test to classify each critical point as a local maximum, local minimum, or neither.

5. Consider the function $g(x) = 3x^5 - 5x^4 + 2$.
   (a) Find the open intervals on which the graph of $g(x)$ is concave up and concave down. Show your work clearly using a sign chart for $g''(x)$.
   (b) Find all inflection points of $g(x)$, providing both x and y coordinates.
   (c) Use the Second Derivative Test, where applicable, to re-verify whether the critical point at $x = 1$ corresponds to a local maximum or local minimum.